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INTERDIGITATED CANTILEVER LEAD FRAME IN CHIP-ON-LEAD PACKAGE DESIGN

Abstract

A semiconductor package has a chip-on-lead configuration with leads that are cantilevered and interdigitated. The leads extend from contact surfaces on opposite sides of the semiconductor package to support locations under a semiconductor die that is attached to the leads. The contact surfaces are exposed at a bottom surface of the semiconductor package, adjacent to a perimeter of the semiconductor package. The leads from opposite sides of the semiconductor package extend past each other under the die. The cantilevered leads do not extend to the bottom surface under the semiconductor die. The semiconductor package includes electrical connections from the semiconductor die to the leads. The semiconductor package is formed by attaching the semiconductor die to the leads, followed by forming the electrical connections.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates to the field of semiconductor packages. More particularly, but not exclusively, this disclosure relates to chip-on-lead semiconductor packages.

BACKGROUND

[0002] Chip-on-lead quad flat no lead (QFN) semiconductor packages have semiconductor die attached to leads, rather than die pads. The chip-on-lead configuration provides smaller footprints compared to packages with die pads.

SUMMARY

[0003] The present disclosure introduces a semiconductor package having a chip-on-lead configuration with two leads that are cantilevered and interdigitated. The leads extend from contact surfaces on opposite sides of the semiconductor package. A semiconductor die is attached to the leads. The leads extend past each other under the die. The leads are not exposed at a bottom surface of the semiconductor package under the semiconductor die. The semiconductor package includes electrical connections from the semiconductor die to each of the leads. The semiconductor package is formed by attaching the semiconductor die to the leads. The contact surfaces extend downward farther than the leads under the semiconductor die. The electrical connections are formed after the semiconductor die is attached to the leads.

Description

BRIEF DESCRIPTION OF THE VIEWS OF THE DRAWINGS

[0004] FIG. 1A through FIG. 1I depict an example semiconductor package having cantilevered interdigitated leads, in stages of an example method of formation.

[0005] FIG. 2A through FIG. 2G depict another example semiconductor package having cantilevered interdigitated leads, in stages of another example method of formation.

[0006] FIG. 3A, a top view, and FIG. 3B, a cross section, depict a further example semiconductor package having cantilevered interdigitated leads.

DETAILED DESCRIPTION

[0007] The present disclosure is described with reference to the attached figures. The figures are not drawn to scale and they are provided merely to illustrate the disclosure. Several aspects of the disclosure are described below with reference to example applications for illustration. It should be understood that numerous specific details, relationships, and methods are set forth to provide an understanding of the disclosure. The present disclosure is not limited by the illustrated ordering of acts or events, as some acts may occur in different orders and/or concurrently with other acts or events. Furthermore, not all illustrated acts or events are required to implement a methodology in accordance with the present disclosure.

[0008] A semiconductor package includes a semiconductor die and has a chip-on-lead configuration, that is, the semiconductor die partially overlaps leads of the semiconductor package and is attached to the leads through a die attach material. The leads include a first lead that extends from a first contact surface adjacent to a first side of the semiconductor package to a first support location under the semiconductor die. The leads include a second lead that extends from a second contact surface adjacent to a second side of the semiconductor package, opposite from the first side, to a second support location under the semiconductor die. The first contact surface and the

second contact surface are exposed through an encapsulation material, which is electrically non-conductive, at a bottom surface of the semiconductor package.

[0009] The first lead and the second lead are cantilevered, that is, the first lead and the second lead do not extend to a bottom surface of the semiconductor package under the semiconductor die. The first contact surface does not extend under the semiconductor die; the second contact surface does not extend under the semiconductor die. The cantilevered configuration, with electrically non-conductive encapsulation material under cantilevered portions of the leads, may advantageously enable placement of the semiconductor package on a circuit substrate, such as a printed circuit board, with conductors between the contact surfaces.

[0010] The two leads are interdigitated, that is, the first lead extends past the second lead under the semiconductor die. In terms of the support locations, a distance between the first support location and the first side of the semiconductor package is greater than a distance between the second support location and the first side of the semiconductor package. Correspondingly, a distance between the second support location and the second side of the semiconductor package is greater than a distance between the first support location and the second side of the semiconductor package.

[0011] The semiconductor package includes electrical connections from the semiconductor die to each of the first lead and the second lead. The electrical connections may be implemented as wire bonds. The chip-on-lead configuration may advantageously enable a smaller area for the semiconductor package by enabling the electrical connections at the leads to be placed closer to the semiconductor die than would be practical for a semiconductor package having a die pad.

[0012] The interdigitated configuration of the first lead and the second lead may support the semiconductor die during formation of the electrical connections at the semiconductor die to provide more reliability for the semiconductor package, compared to packages with cantilevered leads that are not interdigitated.

[0013] FIG. 1A through FIG. 1I depict an example semiconductor package having cantilevered interdigitated leads, in stages of an example method of formation. FIG. 1A is a top perspective view and FIG. 1B is a bottom perspective view of the leads of the semiconductor package. Referring to FIG. 1A and FIG. 1B, the semiconductor package **100** includes leads **102**. The leads **102** may be parts of a lead frame that has additional leads for additional semiconductor packages, not specifically shown. The leads **102** include a first lead **102a** and a second lead **102b**. Each of the leads **102** includes a contact surface **104** at a bottom of the corresponding lead **102**. The first lead has a first contact surface **104a**, and the second lead **102b** has a second contact surface **104b**. The contact surfaces **104** are located in a plane lower than the rest of the leads **102**.

[0014] Referring to FIG. 1C, which is a top perspective view, a first die attach part **106a** of a die attach material **106** is formed on the first lead **102a** and the second lead **102b**. In this example, the first die attach part **106a** may be implemented as die attach paste **106a**. The die attach paste **106a** may include epoxy, by way of example. The first die attach part **106a** may be formed on the first lead **102a** and the second lead **102b** by a dispense process, by way of example. The first die attach part **106a** may include metal particles, such as silver particles or nickel-plated copper particles, which may advantageously improve adhesion to the first and second leads **102a** and **102b**, compared to an unfilled die attach paste.

[0015] Referring to FIG. 1D, a semiconductor die **108** and a second die attach part **106b** of the die attach material **106** are attached to the leads **102** through the first die attach part **106a** of the die attach material **106**. In this example, the second die attach part **106b** may be implemented as a die attach film **106b** attached to a bottom surface of the semiconductor die **108**. The die attach film **106b** may advantageously improve adhesion to the semiconductor die **108**, compared to die attach paste. The die attach film **106b** may be singulated with the semiconductor die **108**, so that a perimeter of the die attach film **106b** is coterminous with a perimeter of the semiconductor die **108**, as depicted in FIG. 1D. Alternatively, die attach part **106a** or die attach film **106b** may be omitted.

[0016] The semiconductor die **108** may be implemented as an integrated circuit, a discrete transistor, a micro electrical mechanical system (MEMS) device, an electro-optical device, or a microfluidic device, by way of example. Other manifestations of the semiconductor die **108** are within the scope of this example. The semiconductor die **108** partially overlaps the first and second leads **102a** and **102b**, providing a chip-on-lead configuration for the semiconductor package **100**. The semiconductor die **108** may include bond pads **108a** on a top surface, opposite from the bottom surface.

[0017] Referring to FIG. 1E, which is a top view of the semiconductor package **100**, the first lead **102a** extends from the first contact surface **104a**, adjacent to a first side **110a** of the semiconductor package **100**, to first support locations **112a** under the semiconductor die **108**. The semiconductor die **108** is attached to the first lead **102a** at the first support locations **112a** by the die attach material **106**. The second lead **102b** extends from the second contact surface **104b**, adjacent to a second side **110b** of the semiconductor package **100**, opposite from the first side, to a second support location **112b** under the semiconductor die **108**. The semiconductor die **108** is attached to the second lead **102b** at the second support location **112b** by the die attach material **106**. In this example, the first lead **102a** is thicker over the first contact surface **104a** than under the semiconductor die **108**, and the second lead **102b** is thicker over the second contact surface **104b** than under the semiconductor die **108**. The semiconductor die **108** is located over cantilevered portions of the first lead **102a** and the second lead **102b**.

[0018] The first lead **102a** and the second lead **102b** are interdigitated, that is, the first lead **102a** extends past the second lead **102b** under the semiconductor die. In one aspect, a first distance **114a** between the first support locations **112a** and the first side **110a** of the semiconductor package **100** may be greater than a second distance **114b** between the second support location **112b** and the first side **110a** of the semiconductor package **100**. A third distance **114c** between the second support location **112b** and the second side **110b** may be greater than a fourth distance **114d** between the first support locations **112a** and the second side **110b**. In another aspect, the first support locations **112a** may be closer to the second contact surface **104b** than they are to the first contact surface **104a**, and the second support location **112b** may be closer to the first contact surface **104a** than it is to the second contact surface **104b**.

[0019] Referring to FIG. 1F, which is a perspective view of the semiconductor package **100**, electrical connections **116** are formed between the bond pads **108a** of the semiconductor die **108** and the leads **102**. The electrical connections **116** may be manifested as wire bonds **116**, formed by a wire bond process using a wire bond tool **118**, with ball bonds on the bond pads **108a** and stitch bonds on the leads **102**, as depicted in FIG. 1F. The die attach paste **106a**, obscured in FIG. 1F by the die attach film **106b** and the semiconductor die **108**, may be recessed from the perimeter of the semiconductor die **108** to allow for resin bleed on the leads **102**, enabling placement of the stitch bonds close to the semiconductor die **108**, advantageously providing a reduced area of the semiconductor package **100**. The wire bond process may be implemented as an ultrasonic wire bond process, a thermocompression wire bond process, or an ultrasonic thermocompression wire bond process, by way of example. Other types of electrical connections, such as ribbon bonds or clip bonds, are within the scope of this example. In this example, the first lead **102a** and the second lead **102b** are not contacted directly below the semiconductor die **108** by elements of the wire bond tool **118** forming the electrical connections **116**. The interdigitated configuration of the first lead **102a** and the second lead **102b** may advantageously provide more support to the semiconductor die **108** during formation of the electrical connections **116** to the bond pads **108a**.

[0020] Referring to FIG. 1G, which is a top view of the semiconductor package **100**, the electrical connections **116** to the semiconductor die **108** are located over cantilevered portions of the first lead **102a** and the second lead **102b**. The electrical connections **116** to the semiconductor die **108** may be located directly over a full thickness of the die attach material **106**, including the die attach paste **106a**, which may further improve support to the semiconductor die **108** during formation of

the electrical connections **116** to the bond pads **108a**.

[0021] Referring to FIG. 1H and FIG. 1I, which are perspective views of the semiconductor package **100**, an encapsulation material **120** is formed on the leads **102** and the semiconductor die **108**, surrounding the electrical connections **116**. The encapsulation material **120** is electrically non-conductive. The encapsulation material **120** may include epoxy or benzocyclobutene (BCB), by way of example. The encapsulation material **120** may include dielectric particles, such as silicon dioxide or aluminum oxide, to reduce a thermal expansion coefficient of the encapsulation material **120**, reducing stress on the semiconductor die **108**. After the encapsulation material **120** is formed, the semiconductor package **100** is singulated, separating the leads **102** from the lead frame. The semiconductor package **100** may be singulated by a saw process, for example.

[0022] The encapsulation material **120** extends to a bottom surface of the semiconductor package **100**, under the cantilevered portions of the leads **102**. The contact surfaces **104** are exposed through the encapsulation material **120** at the bottom surface of the semiconductor package **100**. The first side **110a** of the semiconductor package **100** is located at the bottom surface of the semiconductor package **100**, adjacent to the first contact surface **104a**. The second side **110b** of the semiconductor package **100** is located at the bottom surface of the semiconductor package **100**, adjacent to the second contact surface **104b**. Having the electrically non-conductive encapsulation material **120** under the cantilevered portions of the leads **102** may advantageously enable use of the semiconductor package **100** on a circuit substrate with conductors between the contact surfaces **104**.

[0023] FIG. 2A through FIG. 2G depict another example semiconductor package having cantilevered interdigitated leads, in stages of another example method of formation. Referring to FIG. 2A, a top perspective view, and FIG. 2B, a bottom perspective view, the semiconductor package **200** includes leads **202**, which may be parts of a lead frame. The leads **202** include a first lead **202a** and a second lead **202b**. Each of the leads **202** includes a contact surface **204** at a bottom of the corresponding lead **202**. The first lead has a first contact surface **204a**, and the second lead **202b** has a second contact surface **204b**. The contact surfaces **204** are located in a plane lower than the rest of the leads **202**. The leads **202** of this example may be stamped from a sheet of copper, so that the top surfaces of the leads **202** are indented over the contact surfaces **204**, as depicted in FIG. 2A and FIG. 2B.

[0024] A die attach material **206** is formed on the first lead **202a** and the second lead **202b**. In this example, the die attach material **206** may be implemented as die attach paste **206**. The die attach paste **206** may have a composition as disclosed in reference to the die attach paste **106a** of FIG. 1C.

[0025] FIG. 2C is a cross section of the second lead **202b** through the second contact surface **204b**. The second lead **202b** has a top level thickness **222a** extending to a top surface of the second lead **202b**, and has a contact level thickness **222b** over the second contact surface **204b**. The top level thickness **222a** may be substantially equal to the contact level thickness **222b**, within tolerances encountered in stamping sheet metal, as a result of forming the leads **202** by stamping.

[0026] Referring to FIG. 2D, which is a perspective view, a semiconductor die **208** is attached to the first lead **202a** and the second lead **202b** through the die attach material **206**. The die attach material **206** is subsequently cured. Using the die attach paste **206** may advantageously reduce a cost of the semiconductor package **200** compared to a more complex die attach material **206**.

[0027] Referring to FIG. 2E, which is a perspective view, electrical connections **216** are formed between the semiconductor die **208** and the leads **202**. The electrical connections **216** may be manifested as wire bonds **216**. The interdigitated configuration of the first lead **202a** and the second lead **202b** may advantageously provide more support to the semiconductor die **208** during formation of the electrical connections **216** to the semiconductor die **208**.

[0028] Referring to FIG. 2F, which is a top view of the semiconductor package **200**, the first lead **202a** extends from the first contact surface **204a**, adjacent to a first side **210a** of the semiconductor package **200**, to a first support location **212a** under the semiconductor die **208**. The semiconductor

die **208** is attached to the first lead **202a** at the first support location **212a** through the die attach material **206**. The second lead **202b** extends from a second contact surface **204b**, adjacent to a second side **210b** of the semiconductor package **200**, opposite from the first side, to a second support location **212b** under the semiconductor die **208**. The semiconductor die **208** is attached to the second lead **202b** at the second support location **212b** through the die attach material **206**. The semiconductor die **208** is located over cantilevered portions of the first lead **202a** and the second lead **202b**.

[0029] The first lead **202a** and the second lead **202b** are interdigitated, that is, the first lead **202a** extends past the second lead **202b** under the semiconductor die **208**. In one aspect, a first distance **214a** between the first support location **212a** and the first side **210a** of the semiconductor package **200** may be greater than a second distance **214b** between the second support location **212b** and the first side **210a** of the semiconductor package **200**. A third distance **214c** between the second support location **212b** and the second side **210b** may be greater than a fourth distance **214d** between the first support locations **212a** and the second side **210b**. In another aspect, the first support location **212a** may be closer to the second contact surface **204b** than it is to the first contact surface **204a**, and the second support location **212b** may be closer to the first contact surface **204a** than it is to the second contact surface **204b**.

[0030] The electrical connections **216** to the semiconductor die **208** are located over cantilevered portions of the first lead **202a** and the second lead **202b**. The electrical connections **216** to the semiconductor die **208** may be located directly over the die attach material **206**, which may further improve support to the semiconductor die **208** during formation of the electrical connections **216** to the semiconductor die **208**.

[0031] Referring to FIG. 2G, which is a perspective view of the semiconductor package **200**, an encapsulation material **220**, which is electrically non-conductive, is formed on the leads **202** and the semiconductor die **208**, surrounding the electrical connections **216**. After the encapsulation material **220** is formed, the semiconductor package **200** is singulated, separating the leads **202** from the lead frame. The encapsulation material **220** extends to a bottom surface of the semiconductor package **200**, under the cantilevered portions of the leads **202**. The contact surfaces **204** are exposed through the encapsulation material **220** at the bottom surface of the semiconductor package **200**. The first side **210a** of the semiconductor package **200** is located at the bottom surface of the semiconductor package **200**, adjacent to the first contact surface **204a**. The second side **210b** of the semiconductor package **200** is located at the bottom surface of the semiconductor package **200**, adjacent to the second contact surface **204b**.

[0032] FIG. 3A, a top view, and FIG. 3B, a cross section, depict a further example semiconductor package having cantilevered interdigitated leads. Referring to FIG. 3A and FIG. 3B, the semiconductor package **300** of this example includes a first lead **302a**, a second lead **302b**, a third lead **302c**, and a fourth lead **302d**, which may be parts of a lead frame. The first lead **302a** has a first contact surface **304a**, the second lead **302b** has a second contact surface **304b**, the third lead **302c** has a third contact surface **304c**, the fourth lead **302d** has a fourth contact surface **304d**. The contact surfaces **304a** through **304d** are located in a plane lower than the rest of the leads **302a** through **302d**. The leads **302a** through **302d** of this example may be etched from a sheet of copper, as indicated in FIG. 3B.

[0033] A semiconductor die **308** is attached to the leads **302a** through **302d** through a die attach material **306**. In this example, the die attach material **306** may be implemented as a die attach film **306** attached to a bottom surface of the semiconductor die **308**. The die attach film **306** may advantageously provide improved adhesion to the semiconductor die **308**. A perimeter of the die attach film **106b** may be coterminous with the semiconductor die **308**, as depicted in FIG. 3A and FIG. 3B.

[0034] The first lead **302a** extends from the first contact surface **304a** to a first support location **312a** under the semiconductor die **308**, and the second lead **302b** extends from the second contact

surface **304b**, opposite from the first contact surface **304a**, to a second support location **312b** under the semiconductor die **308**. Similarly, the third lead **302c** extends from the third contact surface **304c** to a third support location **312c** under the semiconductor die **308**, and the fourth lead **302d** extends from the fourth contact surface **304d**, opposite from the third contact surface **304c**, to a fourth support location **312d** under the semiconductor die **308**. The semiconductor die **308** is attached to each of the leads **302a** through **302d** at the corresponding support locations **312a** through **312d**, respectively, by the die attach material **306**.

[0035] The semiconductor die **308** is located over cantilevered portions of the leads **302a** through **302d**. The first lead **302a** and the second lead **302b** are interdigitated, that is, the first lead **302a** extends past the second lead **302b** under the semiconductor die **308**. The first support location **312a** may be closer to the second contact surface **304b** than it is to the first contact surface **304a**, and the second support location **312b** may be closer to the first contact surface **304a** than it is to the second contact surface **304b**. Similarly, the third lead **302c** and the fourth lead **302d** are interdigitated, that is, the third lead **302c** extends past the fourth lead **302d** under the semiconductor die **308**. The third support location **312c** may be closer to the fourth contact surface **304d** than it is to the third contact surface **304c**, and the fourth support location **312d** may be closer to the third contact surface **304c** than it is to the fourth contact surface **304d**.

[0036] Electrical connections **316** are formed between the semiconductor die **308** and the leads **302a** through **302d**. The electrical connections **316** may be manifested as wire bonds **316**. The interdigitated configuration of the leads **302a** through **302d** may advantageously provide more support to the semiconductor die **308** during formation of the electrical connections **316**. The die attach film **306** may reduce resin bleed compared to die past, enabling the electrical connections **316** to be placed closer to the semiconductor die **308**, advantageously providing a smaller area for the semiconductor package **300**.

[0037] An encapsulation material **320**, which is electrically non-conductive, is formed on the leads **302a** through **302d** and the semiconductor die **308**, surrounding the electrical connections **316**. After the encapsulation material **320** is formed, the semiconductor package **300** is singulated, separating the leads **302a** through **302d** from the lead frame. The encapsulation material **320** extends to a bottom surface of the semiconductor package **300**, under the cantilevered portions of the leads **302a** through **302d**. The contact surfaces **304a** through **304d** are exposed through the encapsulation material **320** at the bottom surface of the semiconductor package **300**.

[0038] Various features of the examples disclosed herein may be combined in other manifestations of example semiconductor packages. For example, any of the semiconductor packages disclosed in reference to FIG. 1A through FIG. 1H, FIG. 2A through FIG. 2G, and FIG. 3A and FIG. 3B may have die attach material including die attach paste, die attach film, or a combination of die attach paste and die attach film. Any of the semiconductor packages disclosed in reference to FIG. 1A through FIG. 1H, FIG. 2A through FIG. 2G, and FIG. 3A and FIG. 3B may have leads that are thicker over the contact surfaces than in the cantilevered portions. Any of the semiconductor packages disclosed in reference to FIG. 1A through FIG. 1H, FIG. 2A through FIG. 2G, and FIG. 3A and FIG. 3B may have leads that have substantially equal thicknesses over the contact surfaces and in the cantilevered portions. Any of the semiconductor packages disclosed in reference to FIG. 1A through FIG. 1H, FIG. 2A through FIG. 2G, and FIG. 3A and FIG. 3B may have one pair of cantilevered interdigitated leads, or more than one pair of cantilevered interdigitated leads.

[0039] While various embodiments of the present disclosure have been described above, it should be understood that they have been presented by way of example only and not limitation. Numerous changes to the disclosed embodiments can be made in accordance with the disclosure herein without departing from the spirit or scope of the disclosure. Thus, the breadth and scope of the present disclosure should not be limited by any of the above described embodiments. Rather, the scope of the disclosure should be defined in accordance with the following claims and equivalents.

Claims

1. A semiconductor package, comprising: a semiconductor die; a first lead, extending from a first contact surface to a first support location under the semiconductor die, wherein: the semiconductor die is attached to the first lead at the first support location; the first contact surface is exposed at a bottom surface of the semiconductor package; and the first lead is not exposed at the bottom surface of the semiconductor package under the semiconductor die; a second lead, extending from a second contact surface to a second support location under the semiconductor die, wherein: the semiconductor die is attached to the second lead at the second support location; the second contact surface is exposed at the bottom surface of the semiconductor package, opposite from the first contact surface; the second lead is not exposed at the bottom surface of the semiconductor package under the semiconductor die; and the first lead and the second lead extend past each other under the semiconductor die; a first electrical connection between the semiconductor die and the first lead; and a second electrical connection between the semiconductor die and the second lead.
2. The semiconductor package of claim 1, wherein: the first contact surface does not extend under the semiconductor die; and the second contact surface does not extend under the semiconductor die.
3. The semiconductor package of claim 1, wherein the semiconductor die is located over cantilevered portions of the first lead and the second lead.
4. The semiconductor package of claim 1, wherein: the first support location is closer to the second contact surface than the first support location is to the first contact surface; and the second support location is closer to the first contact surface than the second support location is to the second contact surface.
5. The semiconductor package of claim 1, wherein: a first distance between the first support location and a first side of the semiconductor package, adjacent to the first contact surface, is greater than a second distance between the second support location and the first side of the semiconductor package; and a third distance between the second support location and a second side of the semiconductor package, adjacent to the second contact surface, is greater than a fourth distance between the first support location and the second side of the semiconductor package.
6. The semiconductor package of claim 1, wherein the semiconductor die is attached to the first lead and to the second lead by a die attach material selected from the group consisting of a die attach paste, a die attach film, and by a combination of a die attach film and a die attach paste.
7. The semiconductor package of claim 6, wherein the die attach material on the first lead and the die attach material on the second lead extend past each other under the semiconductor die.
8. The semiconductor package of claim 1, wherein the first electrical connection and the second electrical connection are wire bonds.
9. The semiconductor package of claim 1, further including electrically insulating package material between the first lead and the bottom surface of the semiconductor package, under the semiconductor die, and between the second lead and the bottom surface of the semiconductor package, under the semiconductor die.
10. A method of forming a semiconductor package, comprising: attaching a semiconductor die to a first lead and a second lead, the first lead extending from a first contact surface to a first support location under the semiconductor die, the second lead extending from a second contact surface, opposite from the first contact surface, to a second support location under the semiconductor die, wherein the first contact surface and the second contact surface extend downward farther than the first lead and the second lead under the semiconductor die, and the first lead and the second lead extend past each other under the semiconductor die; forming a first electrical connection from the semiconductor die to the first lead; and forming a second electrical connection from the semiconductor die to the second lead.

- 11.** The method of claim 10, wherein: the first contact surface does not extend under the semiconductor die; and the second contact surface does not extend under the semiconductor die.
 - 12.** The method of claim 10, wherein the semiconductor die is located over cantilevered portions of the first lead and the second lead.
 - 13.** The method of claim 10, wherein: the first lead is thicker over the first contact surface than under the semiconductor die; and the second lead is thicker over the second contact surface than under the semiconductor die.
 - 14.** The method of claim 10, wherein: the first lead has a same thickness over the first contact surface and under the semiconductor die; and the second lead has a same thickness over the second contact surface and under the semiconductor die.
 - 15.** The method of claim 10, wherein attaching the semiconductor die to the first lead and to the second lead includes dispensing a die attach paste onto the first lead and the second lead, and subsequently placing the semiconductor die on the die attach paste, followed by curing the die attach paste.
 - 16.** The method of claim 10, wherein attaching the semiconductor die to the first lead and to the second lead includes attaching a die attach film, that is attached to the semiconductor die, to the first lead and to the second lead, followed by curing the die attach film.
 - 17.** The method of claim 10, wherein attaching the semiconductor die to the first lead and to the second lead includes dispensing a die attach paste onto the first lead and the second lead, and subsequently placing a die attach film, that is attached to the semiconductor die, to the first lead and to the second lead, followed by curing the die attach film.
 - 18.** The method of claim 10, wherein forming the first electrical connection includes forming a wire ball bond to the semiconductor die.
 - 19.** The method of claim 18, wherein forming the wire ball bond is performed with a ball bond process selected from the group consisting of a thermocompression ball bond process, an ultrasonic ball bond process, and an ultrasonic thermocompression ball bond process.
 - 20.** The method of claim 10, wherein the first lead and the second lead are not contacted directly below the semiconductor die by equipment forming the first electrical connection and the second electrical connection.
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